

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
“Jnana Sangama”, Belgaum -590014, Karnataka.



PROJECT WORK REPORT
On

“TogetherInCrisis”

Submitted by

ANU SAI SHREE R (1BM23CS045)
ANJANA MANOJ (1BM23CS038)
ARCHITA V (1BM23CS050)
ANAGHA D K (1BM23CS032)

Under the Guidance of
Sandhya A Kulkarni
Assistant Professor

in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING



B.M.S. COLLEGE OF ENGINEERING
(Autonomous Institution under VTU)
BENGALURU-560019
Sept 2024-Jan 2025

**B. M. S. College of Engineering,
Bull Temple Road, Bangalore 560019
(Affiliated To Visvesvaraya Technological University, Belgaum)
Department of Computer Science and Engineering**



CERTIFICATE

This is to certify that the project work entitled “**TogetherInCrisis**” carried out by **ANU SAI SHREE R(1BM23CS045), ANJANA MANOJ (1BM23CS038), ARCHITA V (1BM23CS050), ANAGHA D K (1BM23CS032)** who are bonafide students of **B. M. S. College of Engineering**. It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visveswararajah Technological University, Belgaum during the year 2024. The project report has been approved as it satisfies the academic requirements in respect of **Full Stack Web Development(23CS3AEFWD)** work prescribed for the said degree.

Signature of the Guide

Sandhya A Kulkarni
Assistant Professor
BMSCE, Bengaluru

Signature of the HOD

Dr. Kavitha Sooda
Prof & Head of Dept of CSE
BMSCE, Bengaluru

External Viva

Name of the Examiner

Signature with date

**B.M.S. COLLEGE OF ENGINEERING DEPARTMENT OF COMPUTER
SCIENCE AND ENGINEERING**



DECLARATION

We, **ANU SAI SHREE R(1BM23CS045), ANJANA MANOJ (1BM23CS038), ARCHITA V(1BM23CS050), ANAGHA D K(1BM23CS032)** students of 3rd Semester, B.E, Department of Computer Science and Engineering, BMS College of Engineering, Bangalore, hereby declare that, this Full Stack Web Development entitled "**TOGETHER IN CRISIS**" has been carried out by us under the guidance of Sandhya A Kulkarni, Assistant Professor, Department of CSE, BMS College of Engineering, Bangalore during the academic semester Sept 2024 - Jan 2025.

We also declare that to the best of our knowledge and belief, the development reported here is not from part of any other report by any other students.

Signature

ANU SAI SHREE R (1BM23CS045)
ANJANA MANOJ (1BM23CS038)
ARCHITA V (1BM23CS050)
ANAGHA D K (1BM23CS032)

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1.INTRODUCTION

1.1.Overview

TogetherInCrisis is an innovative platform designed to connect individuals and organizations in times of crisis. It serves as a hub for volunteers, NGOs, and disaster relief agencies to collaborate effectively. The website enables users to offer help and share resources, fostering a coordinated response during emergencies such as natural disasters, pandemics, or humanitarian crises. It aims to streamline communication, resource allocation, and task management in emergency crises.

The website provides features like:

- Volunteer registration and profile management.
- Request submission by individuals or organizations.
- Real-time matching of volunteers to requests based on location and skill set.
- Communication tools for efficient coordination.

The primary objective is to reduce response time, optimize resource distribution, and empower communities to handle crises effectively. This platform serves as a stepping stone toward creating a more resilient and self-sufficient society in the face of unpredictable challenges.

1.2. Motivation

The motivation for this project comes from the need to improve crisis management and make a meaningful impact. Recent events like natural disasters and health emergencies have shown the importance of quick and coordinated volunteer efforts. However, many existing systems face issues such as poor resource allocation, delays in communication, and lack of centralized platforms for organizing volunteers.

This project aims to solve these problems by using modern web technologies to create a platform that is:

Accessible: Easy to use for anyone, regardless of technical skills.

Efficient: Quickly matches volunteers to needs based on location, skills, and availability.

Scalable: Can grow to handle more users and different types of crises.

This project is driven by the belief that every individual has the potential to contribute positively during emergencies, and a robust technological platform can empower communities to respond collectively and effectively. By bridging the gap between those in need and those willing to help, this initiative aims to create a ripple effect of positive change.

2.SOFTWARE REQUIREMENT SPECIFICATION

2.1. Hardware Requirements:

- A PC with the following or greater specifications: ○ Intel Core i5/i7 or AMD Ryzen 5/7 ○ Minimum 8 GB RAM;16 GB is recommended ○ 512 GB SSD
- A stable internet connection (2Mbps or higher)

2.2. Software Requirements:

Operating system : Windows, Linux

Front end technologies are : HTML, CSS, JavaScript

IDE : VS Code,Sublime text

Back end requirement : PHP/Node.js

Server : Express in Node.js

2.3.Web Application

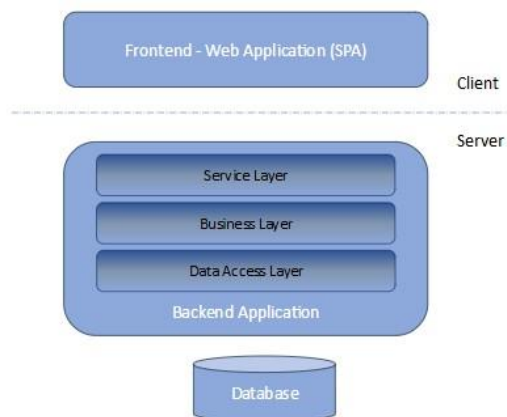


Fig 2.3 Web application of the project

Fig 2.3 illustrates a layered architecture for a web application, featuring:

1. Frontend (SPA): The user interface running in the browser, offering dynamic interaction without full-page reloads.
2. Backend Application: Includes three layers:
 - Data Access Layer: Handles direct interactions with the database.

- Business Layer: Contains the core logic of the application.
- Service Layer: Exposes business logic to the frontend and other clients.

3. Database: Stores persistent data used by the application.

This structure separates concerns for easier maintenance and scalability.

2.4. Technology mapping

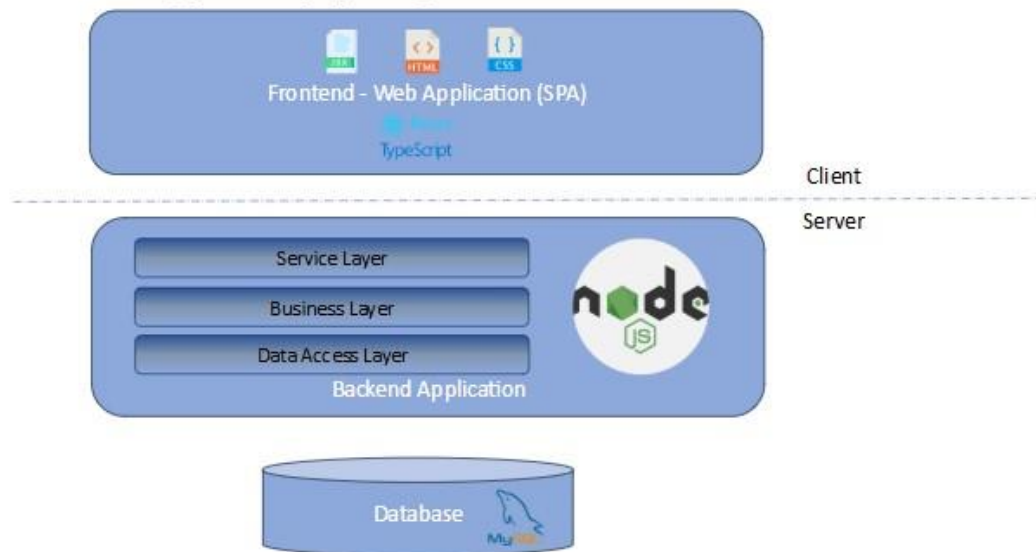


Fig 2.4 Technology mapping for the project

This diagram showcases the architecture of a web application with a Single Page Application (SPA) on the client-side and a Node.js-based backend:

1. Frontend (SPA)-Built using HTML,CSSTypeScript, and React, providing a dynamic, single-page interface for the user.
2. Backend Application
 - Service Layer: Manages communication between the frontend and backend.
 - Business Layer: Handles the core application logic.
 - Data Access Layer: Interacts directly with the database.
- 3.Database: The backend uses MySQL for storing persistent data.

This setup highlights the modern web development stack with Node.js for the server-side and MySQL as the database solution.

3. ER DIAGRAM OF THE PROJECT

The fig 3 shows the following :

A graphical representation of entities and their relationships to each other, typically used in database design. It helps visualize how data is structured and interrelated in a system, providing a high-level blueprint for creating databases. It helps identify the entities, attributes, and relationships clearly and reduces redundancy by ensuring proper normalization. Entities: User, User_type, Testimonials, Events_enrolled, Cause_area, Place, Skill, Event.

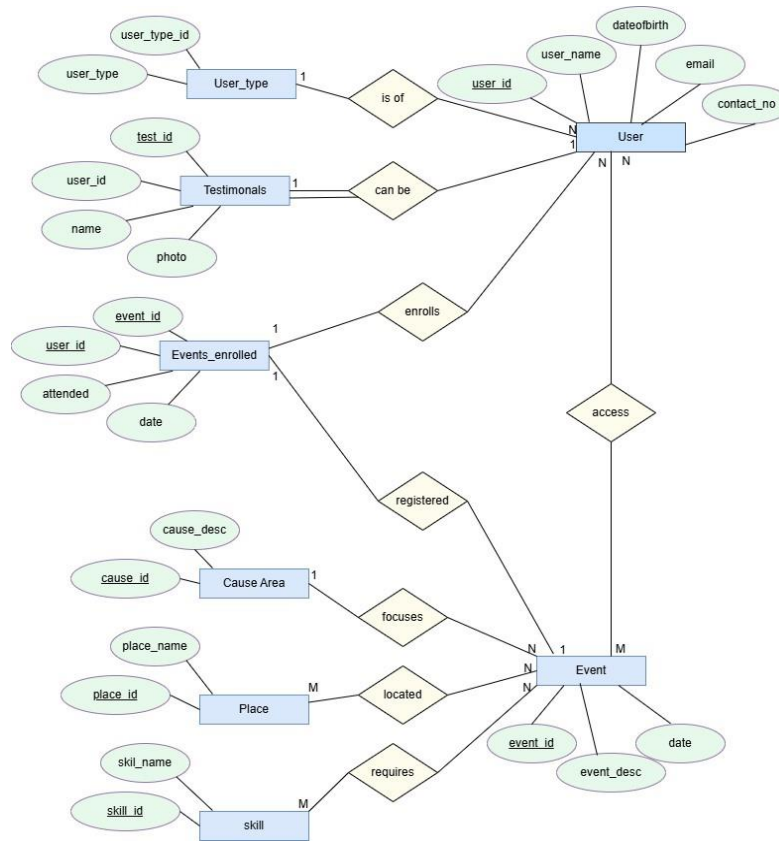


Fig 3 ER Diagram of the project

This is an Entity-Relationship (ER) diagram illustrating the relationships between entities in a user-event management system. The central entity is "User," linked to "User Type," "Testimonials," and "Events Enrolled." Events are described with attributes such as location ("Place"), required skills ("Skill"), and their focus ("Cause Area"). Users enroll in events and may access or register them based on their details. The diagram captures many-to-many and oneto-many relationships, making it suitable for systems tracking users, their activities, and associated event data.

4.Schema Diagram of the Project

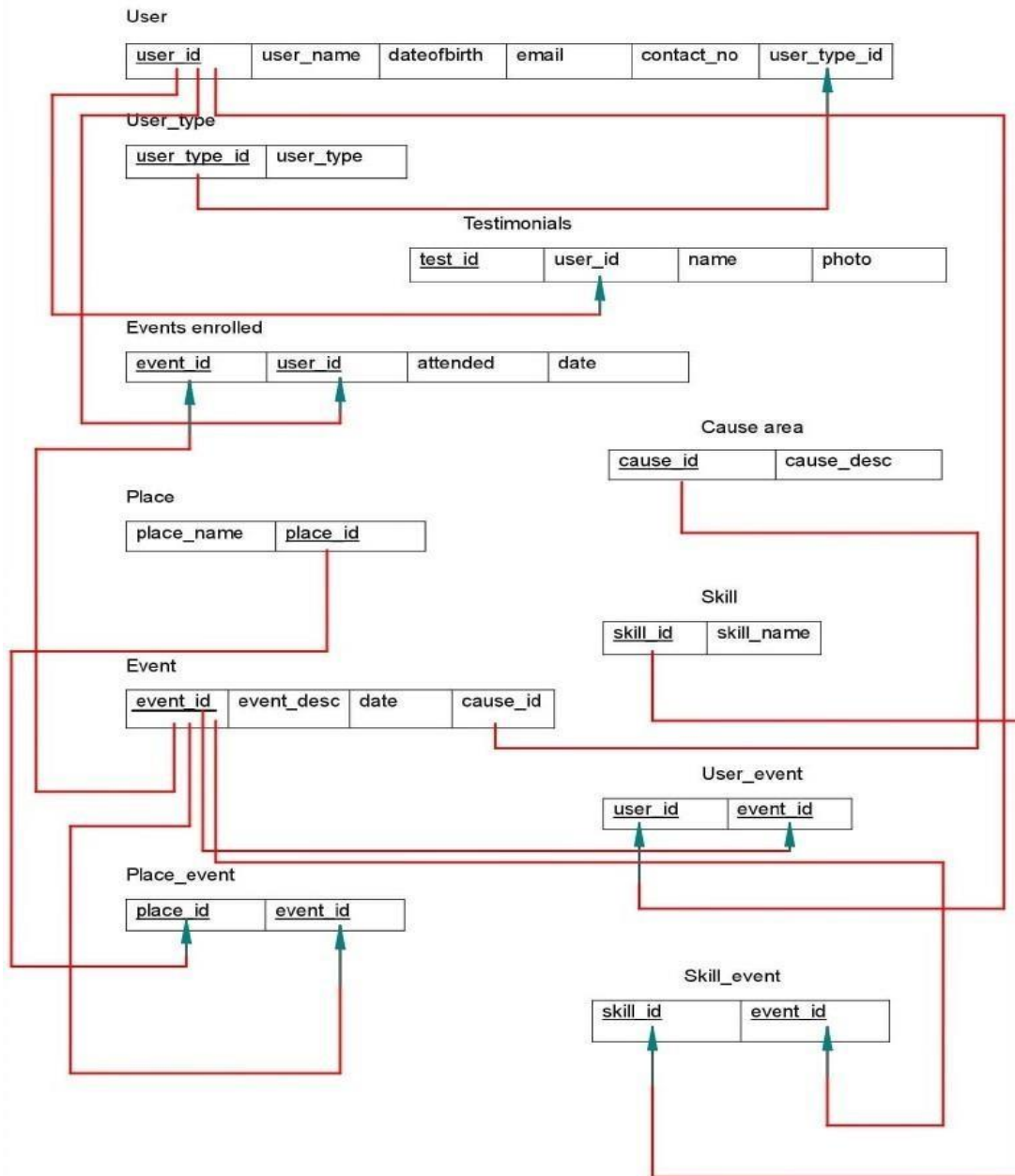


Fig 4 Schema Diagram of the project

Fig 4 illustrates the schema diagram for the required data modelling through E-R Diagram(Fig 3) The entities required and the relationship (cardinality) between the entities are specified by the Fig 4

5.USER INTERFACE DESIGN

5.1.Home page

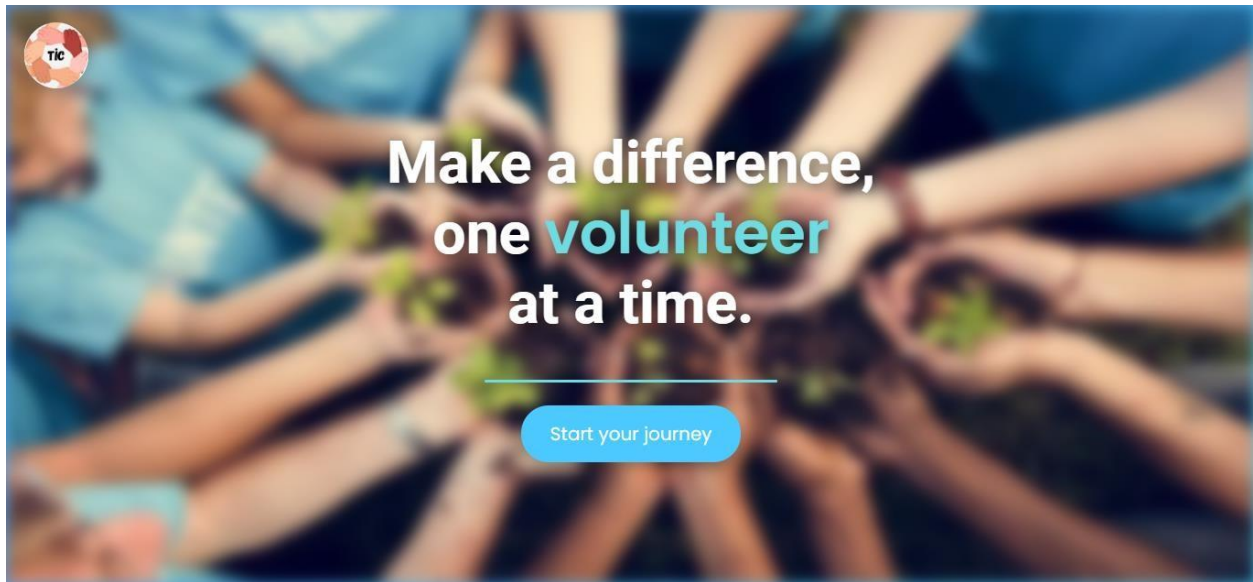


Fig 5.1 Home page

"Make a difference, one volunteer at a time."

Motivates the user to act with an action button "Start your journey." A blurred background image of hands holding small plants represents teamwork, growth, environmental or community concern. The concept is to encourage people to contribute to it by volunteering. It motivates people to contribute in whatever way they can contribute to the betterment of the nature and future of the environment.

This page is designed to motivate and recruit people for volunteering efforts, focusing on the impact of personal involvement in making a difference in society. It appeals to those who are passionate about social or environmental causes.

5.2.Vision and mission

The images present the **vision** and **mission** of our organization "TogetherInCrisis". The vision is centered on creating a compassionate and collaborative world where individuals unite to drive meaningful change. The key focus areas are empowering people to contribute to communities in need, providing opportunities for growth, support, and development regardless of background or circumstances. The mission focuses on building a platform that connects volunteers with impactful opportunities to address critical social, environmental, and cultural challenges. We

cultivate a network of change-makers to foster transformation and a sense of purpose. At the bottom of the page, two prominent buttons provide options to Sign in for existing users or Sign up for new users. The design encourages user engagement by making the next steps simple and accessible.

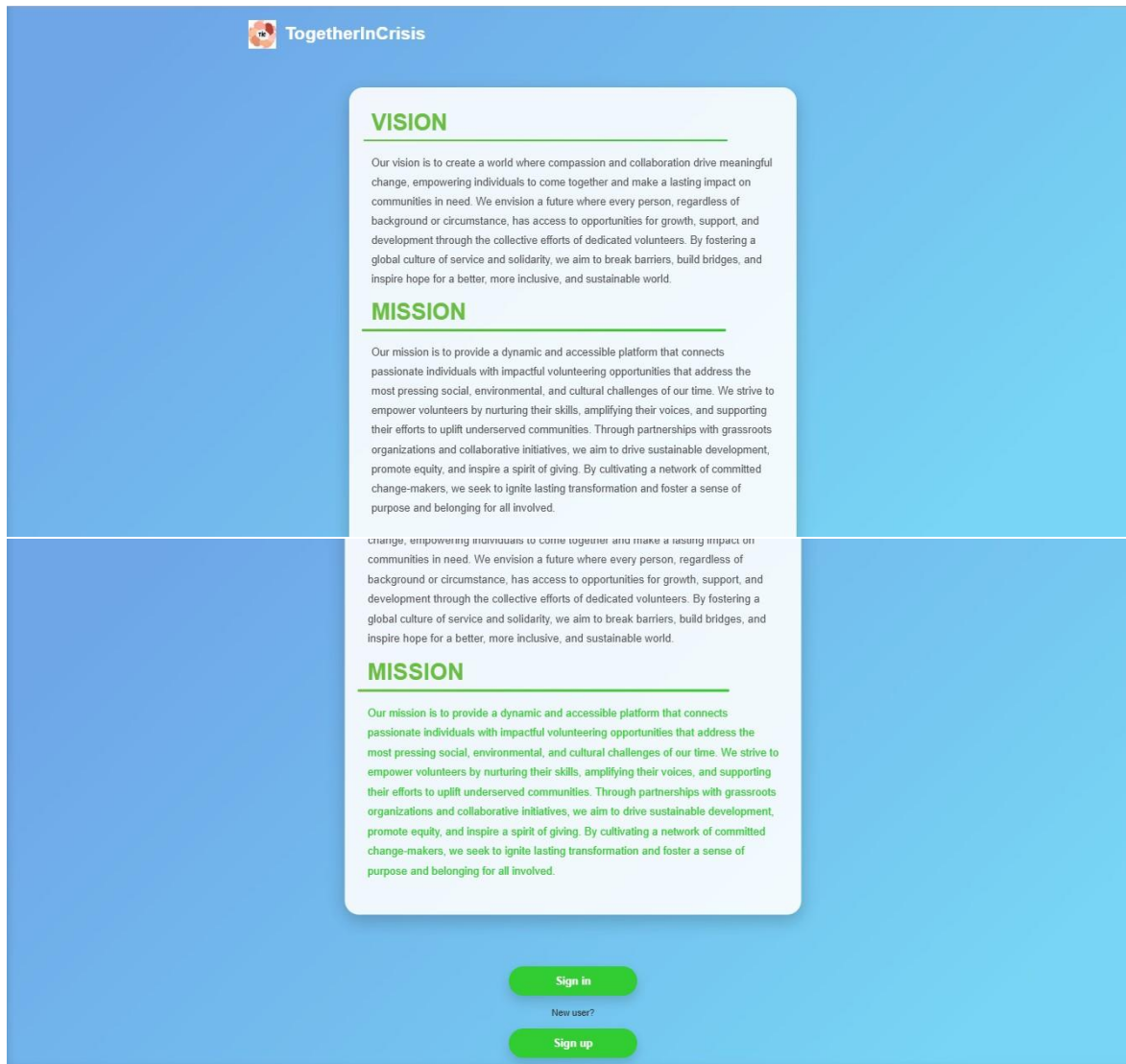


Fig 5.2 Vision and Mission

5.3.Sign-in

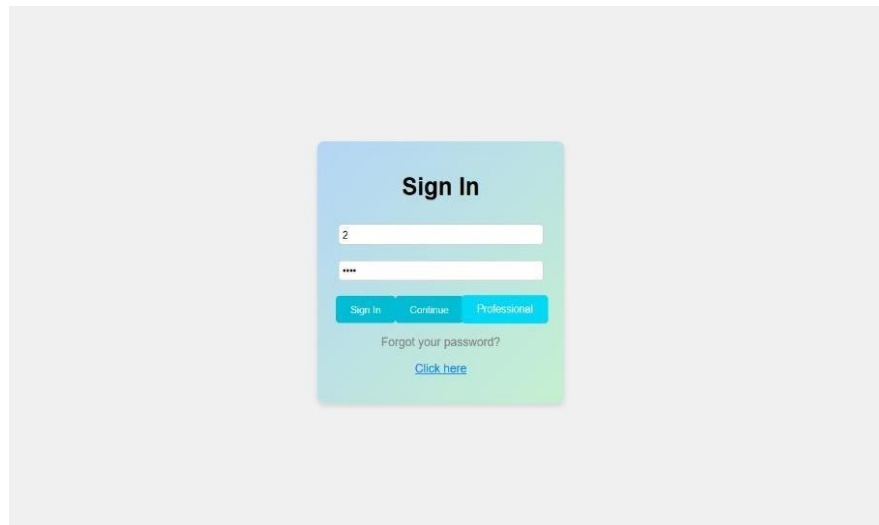


Fig 5.3 Sign-in

The image presents a **Sign In** page. It is for the users who are already registered. We have a separate button for the Professionals who will directly land to their profile page instead of going through the entire process of login. There is a Continue button which will take them to the next landing page.

5.4.Forgot Password

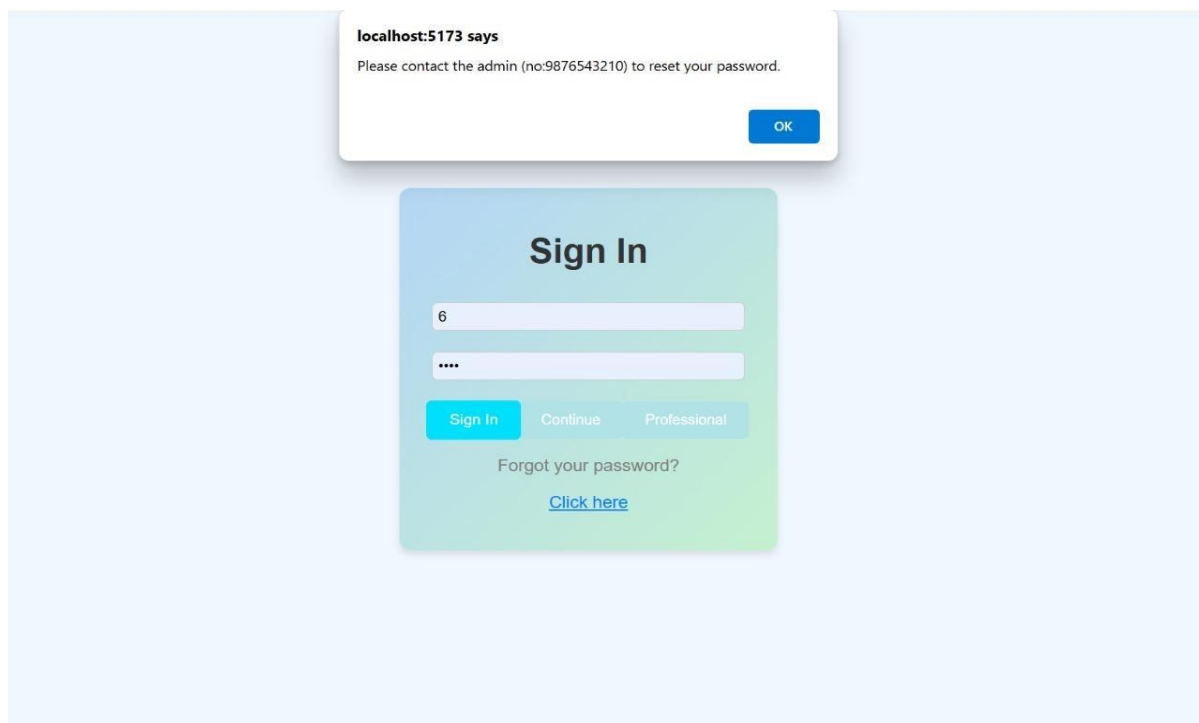


Fig 5.4 Forgot Password

There is a ‘**Forgot password?**’ button. As you click on it, it will redirect you to a page where you will be able contact the administrator for further assistance.

5.5.Sign-up

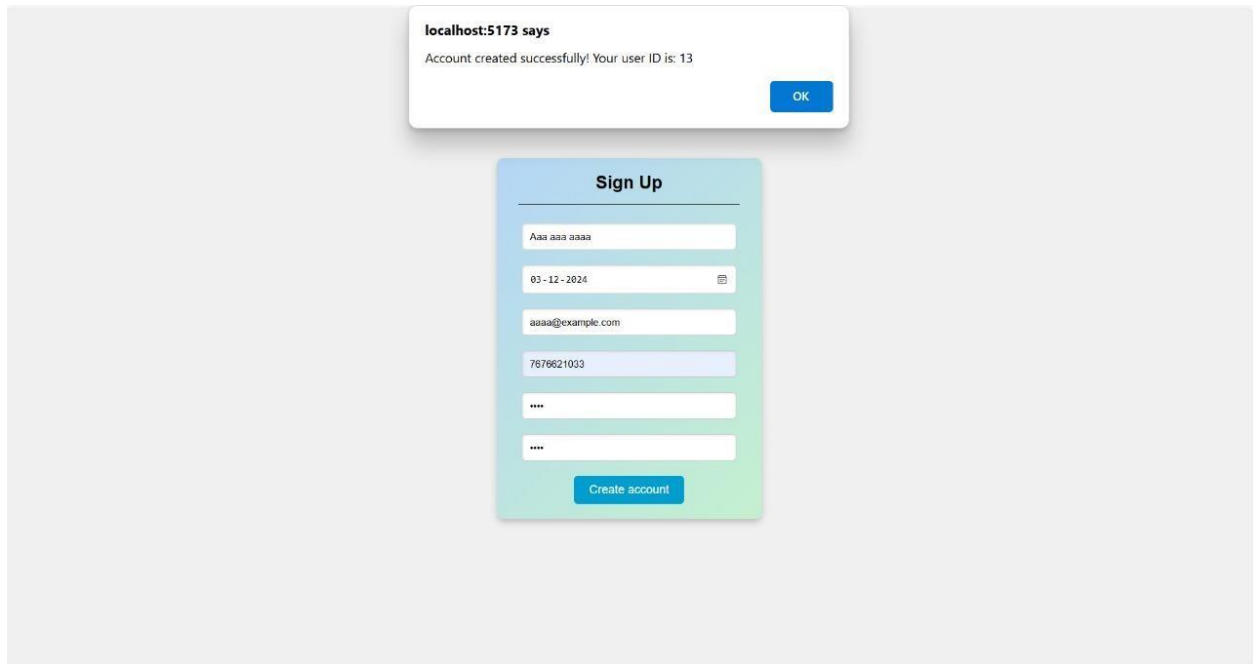
The image shows a web application interface. At the top, a white notification box with a blue border contains the text "localhost:5173 says" and "Account created successfully! Your user ID is: 13", with an "OK" button. Below this is a "Sign Up" form with a light blue header. The form contains several input fields: a name field with "Aaa aaa aaaa", a date field with "03-12-2024", an email field with "aaaa@example.com", a phone number field with "7678621033", and two password fields, each with four asterisks. A blue "Create account" button is at the bottom of the form.

Fig 5.5 Sign-up

The image presents the **Sign-up** page for the users. It is for the new users who are interested in our website. This form is designed to collect user credentials to grant access to the website or application. It helps us to keep a track of the number of users and what more can be done to improve our website.

5.6.Volunteering Type Selection

After signing in or signing up, the user is directed to the Volunteering Type Selection Page, where they are prompted to choose between two main types of volunteering: **Crisis Volunteering** or **Casual Volunteering**.

- Crisis Volunteering involves immediate, high-impact opportunities such as responding to natural disasters, pandemics, or other emergencies.
- Casual Volunteering focuses on ongoing, less urgent tasks like tutoring, teaching, or community-building activities.

This choice helps personalize future event recommendations, ensuring users are matched with volunteering opportunities that align with their interests and skills.



Fig 5.6 Volunteering Type Selection

5.7. Events Page

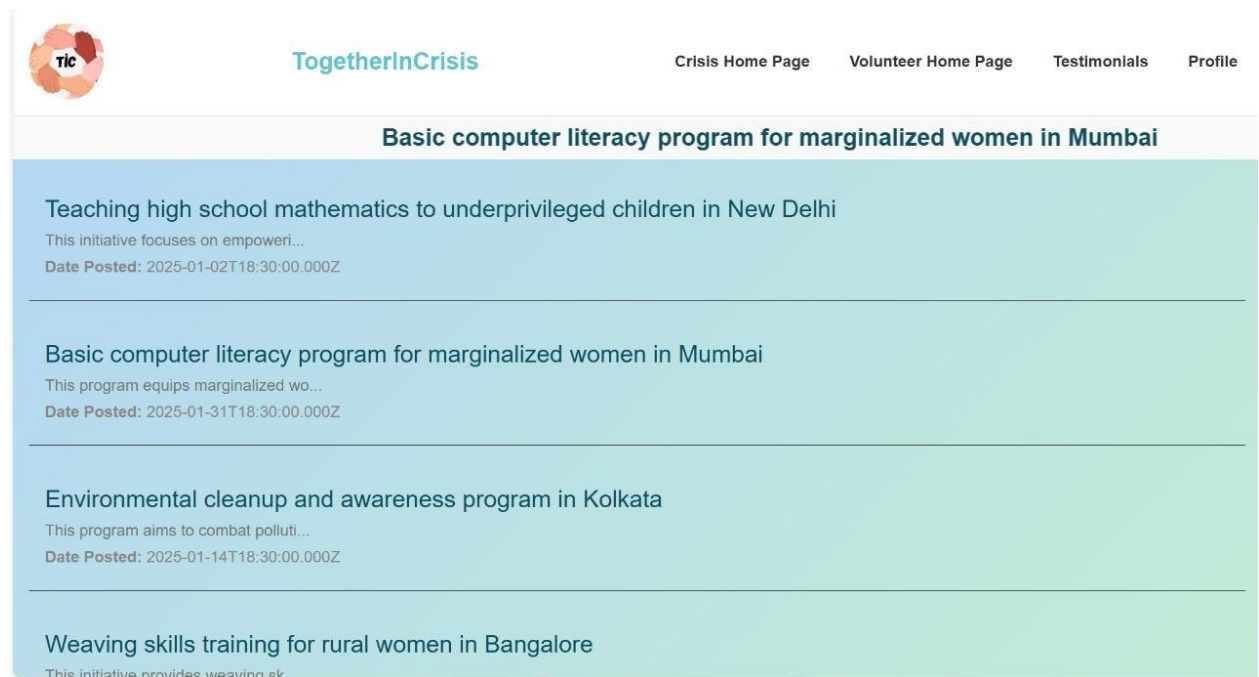


Fig 5.7.1 Volunteering events page

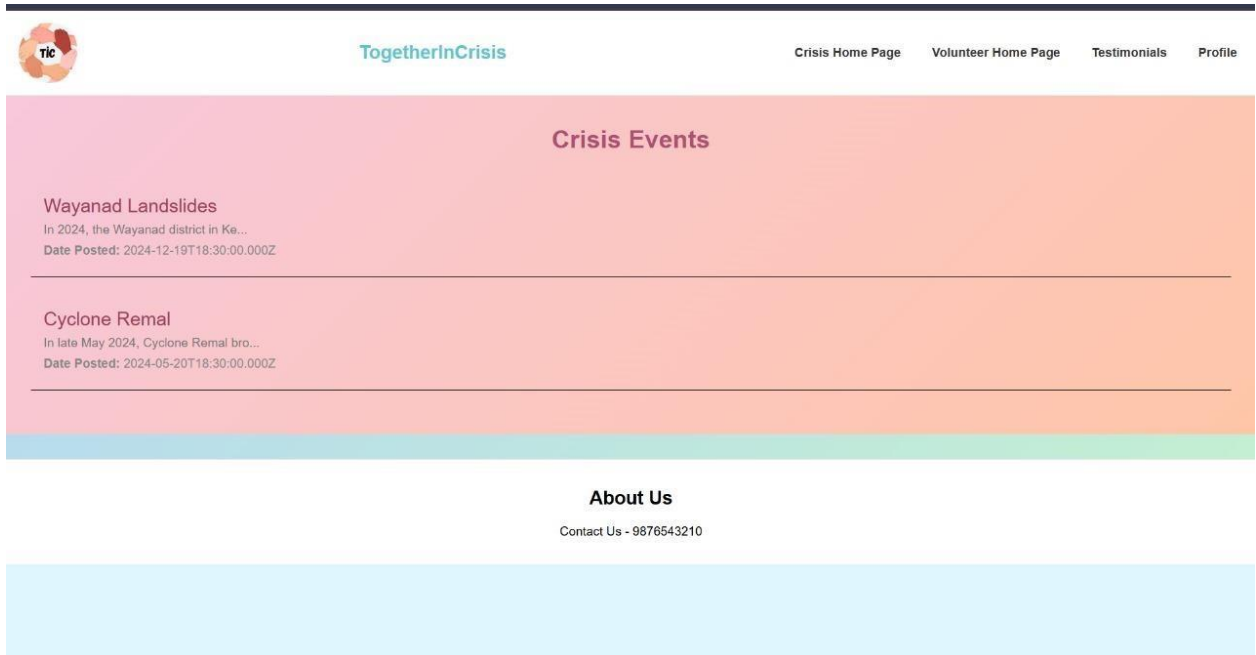
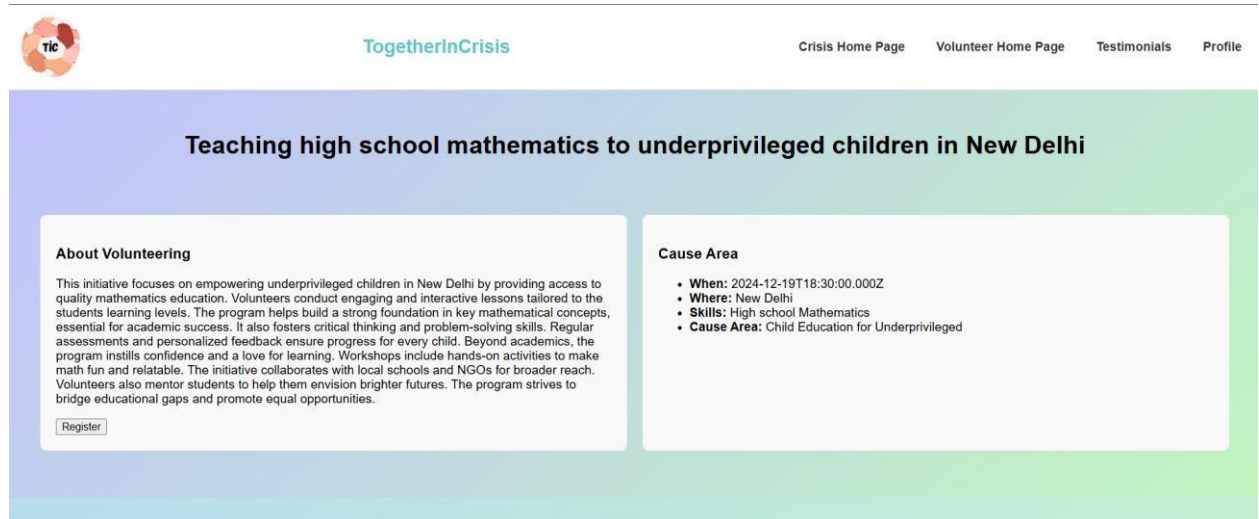


Fig 5.7.2 Crisis events page

- Header:
 - Displays the platform's name/logo ("TogetherInCrisis").
 - Contains a navigation menu with links such as "Crisis Home Page," "Volunteer Home Page," "Testimonials," and "Profile."
- Main Section:
 - Crisis Events: Highlights crisis-related events like:
 - Wayanad Landslides (2024).
 - Cyclone Remal (May 2024).
 - Each event includes a brief description and the posting date.
 - Volunteering events: As shown in Fig 5.7.1 highlights volunteering related events
- Footer: Contains an "About Us" section with contact information.

5.8. Event Details



About Us
Contact Us - 9876543210

Fig 5.8.1 Event details display

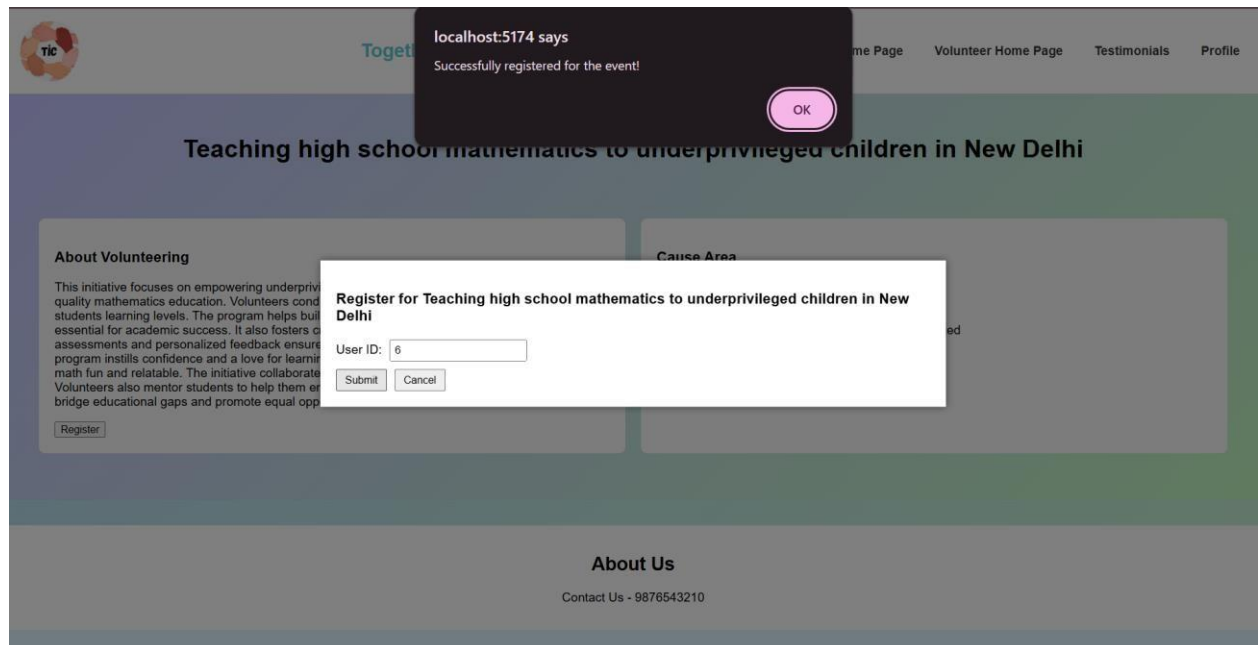


Fig5.8.2 Event registration

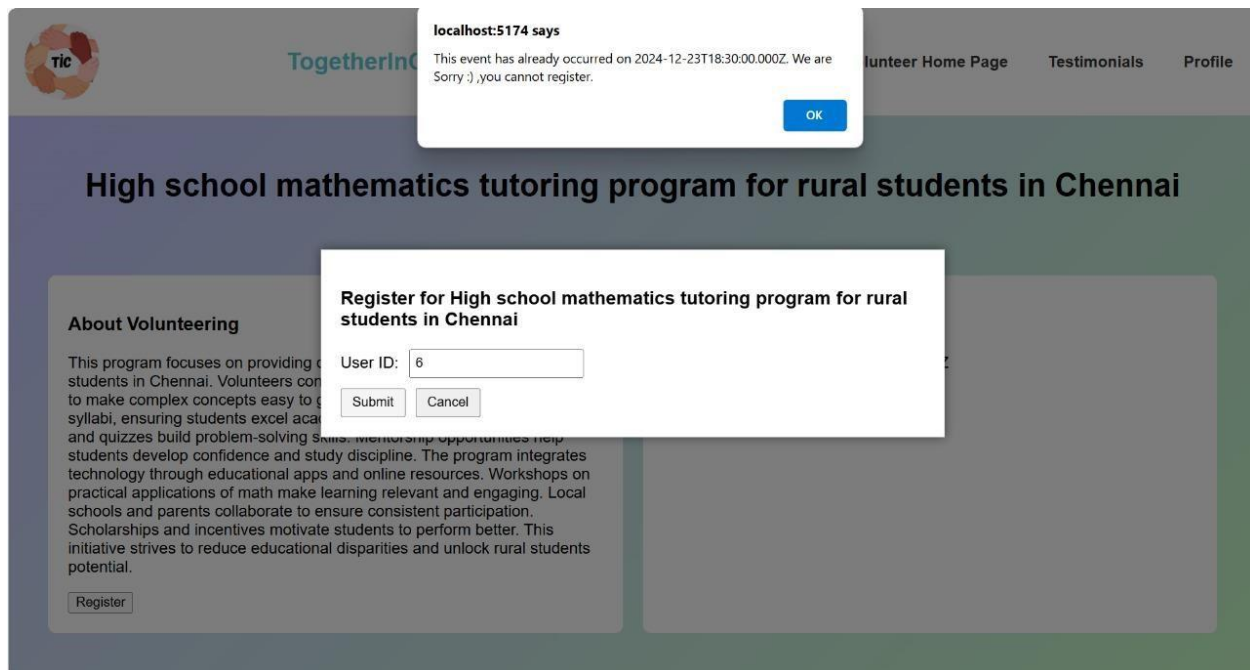


Fig 5.8.3 Event already occurred

Event Details Page:Provides comprehensive information about specific volunteering events, including event name, cause area, schedule, location, date, and requirements.

Features a Register button for easy sign-up.

Highlights the significance of volunteering and its impact on communities.

Encourages user participation by outlining the benefits of contributing to social causes. Displays a contact number at the bottom of the page for easy communication and enquiries. This page helps volunteers to keep track of their events .

5.9. User Profile

The **User Profile Page** offers volunteers a personalized space to manage their account details, including name, email. The page also includes sections for Registered Events, where users can track upcoming commitments, and Participated Events, which showcase past contributions.

With its intuitive design, the User Profile Page promotes ongoing participation and reinforces a sense of accomplishment and community among volunteers


TogetherInCrisis

[Crisis Home Page](#)
[Volunteer Home Page](#)
[Testimonials](#)
[Profile](#)

Events

Date	Event Name	Status
2024-12-28T18:30:00.000Z	Basic computer literacy program for marginalized women in Mumbai	Registered but not attended
2024-12-28T18:30:00.000Z	Environmental cleanup and awareness program in Kolkata	Attended
2024-12-22T18:30:00.000Z	High school mathematics tutoring program for rural students in Chennai	Registered but not attended
2024-12-18T18:30:00.000Z	Wayanad Landslides	Registered but not attended
2024-12-18T18:30:00.000Z	Wayanad Landslides	Registered but not attended
2024-05-19T18:30:00.000Z	Cyclone Remal	Registered but not attended



Anu Sai's Profile

Email: anu@example.com
Contact No: 1234567890
Date of Birth: 2006-03-04T18:30:00.000Z

About Us

Contact Us - 9876543210

Fig 5.9 User Profile

5.10. Professional Profile



Name: Archita V
Email: archiiii@example.com
Phone: 1234567890
Date of Birth: 1980-03-27T18:30:00.000Z

Professional Profile

Volunteer Details	Event	Attendance (Weekly)
Anu Sai	Teaching high school mathematics to underprivileged children in New Delhi	☐
Aarav Sharma	Teaching high school mathematics to underprivileged children in New Delhi	☒
Aarav Sharma	Basic computer literacy program for marginalized women in Mumbai	☐
Anu Sai	Basic computer literacy program for marginalized women in Mumbai	☐
Anu Sai	Environmental cleanup and awareness program in Kolkata	☒
Aarav Sharma	Environmental cleanup and awareness program in Kolkata	☐
Aarav Sharma	Weaving skills training for rural women in Bangalore	☐
Aarav Sharma	Teaching high school mathematics to underprivileged children in New Delhi	☐
Aarav Sharma	High school mathematics tutoring program for rural students in Chennai	☐
Anagha	Wayanad Landslides	☐
Aarav Sharma	Wayanad Landslides	☐
Anu Sai	Wayanad Landslides	☐
Aarav Sharma	Cyclone Remal	☐

Fig 5.10.1 Professional profile

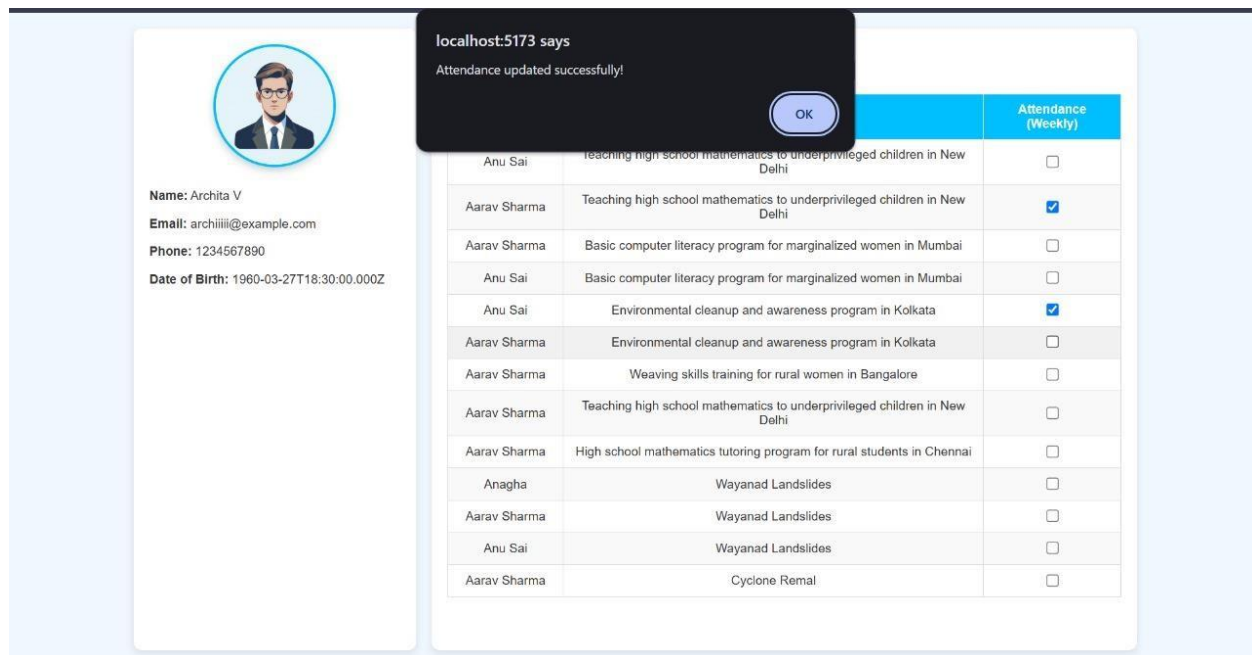


Fig 5.10.2 User attendance updation by professional

The **Professional Profile** page provides a comprehensive summary of a volunteer's details and attendance record. It displays a structured table listing volunteers and their weekly attendance. It also displays personal information of the professional such as name, email, phone number, and status. The page features a clean layout with a calming blue background, ensuring readability and ease of use. It is designed to help coordinators/professionals track volunteer engagement and allow volunteers to review their records. This page serves as a centralized hub for managing volunteer profiles effectively.

This page gets updated weekly or after every event is done. This helps the users to see how their progress is and easy to maintain/keep track of their attendance.

5.11. Testimonials

Our volunteers share their inspiring stories and positive experiences, showcasing the impact and fulfillment they gain from their contributions. These testimonials highlight the sense of community, personal growth, and the meaningful difference our volunteers make in the lives of others. Their feedback underscores the value and significance of volunteering in fostering a

supportive and compassionate environment. This helps the viewers who just come to see the website to get motivated and register as a volunteer.

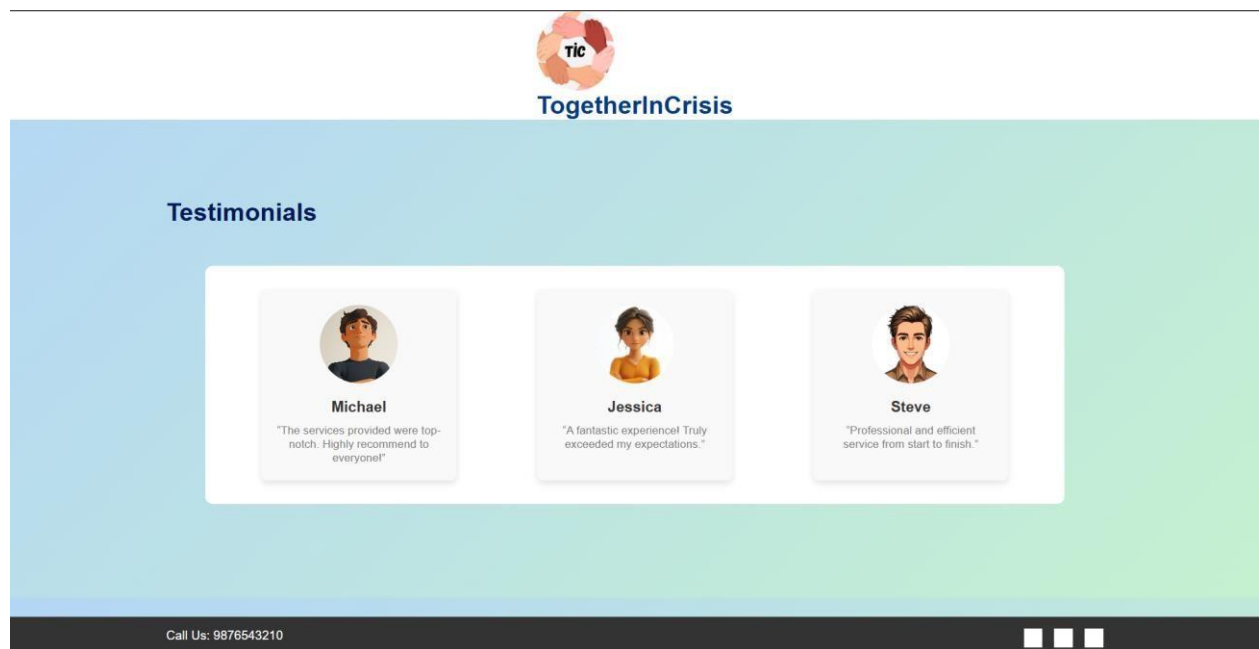


Fig 5.11 Testimonials

REFERENCES:

Tutorial Links:

- [1] <https://www.w3schools.com/html/default.asp>
- [2] <https://nodejs.org/docs/latest/api/>
- [3] <https://expressjs.com/>
- [4] <https://www.postgresql.org/docs/>
- [5] <https://restfulapi.net/>

Referred website:

- [1] <https://www.volunteermatch.org/>
- [2] <https://www.crisistextline.org/>
- [3] <https://www.freecodecamp.org/>

Referred Textbooks:

1. Modern Full Stack Development using typescript , react , nodejs (Author: Frank Zammetti)